

EXPECTATION

- Readings for the next few lectures:

Ch 8 (Watkins)

Ch 3 (Wasserman, for more formal treatment)

REVIEW OF (LAST PART OF) LECTURE 5

- Mean $\equiv$ expected value $\equiv$ expectation $E(X)$ or μ

$$\sum_{\forall x} xf(x), \text{ discrete } X \qquad \int xf(x) dx, \text{ continuous } X$$

- Expectation of a function of a random variable:

$$E[r(X)] = \int r(x)f(x) dx$$

- Properties:

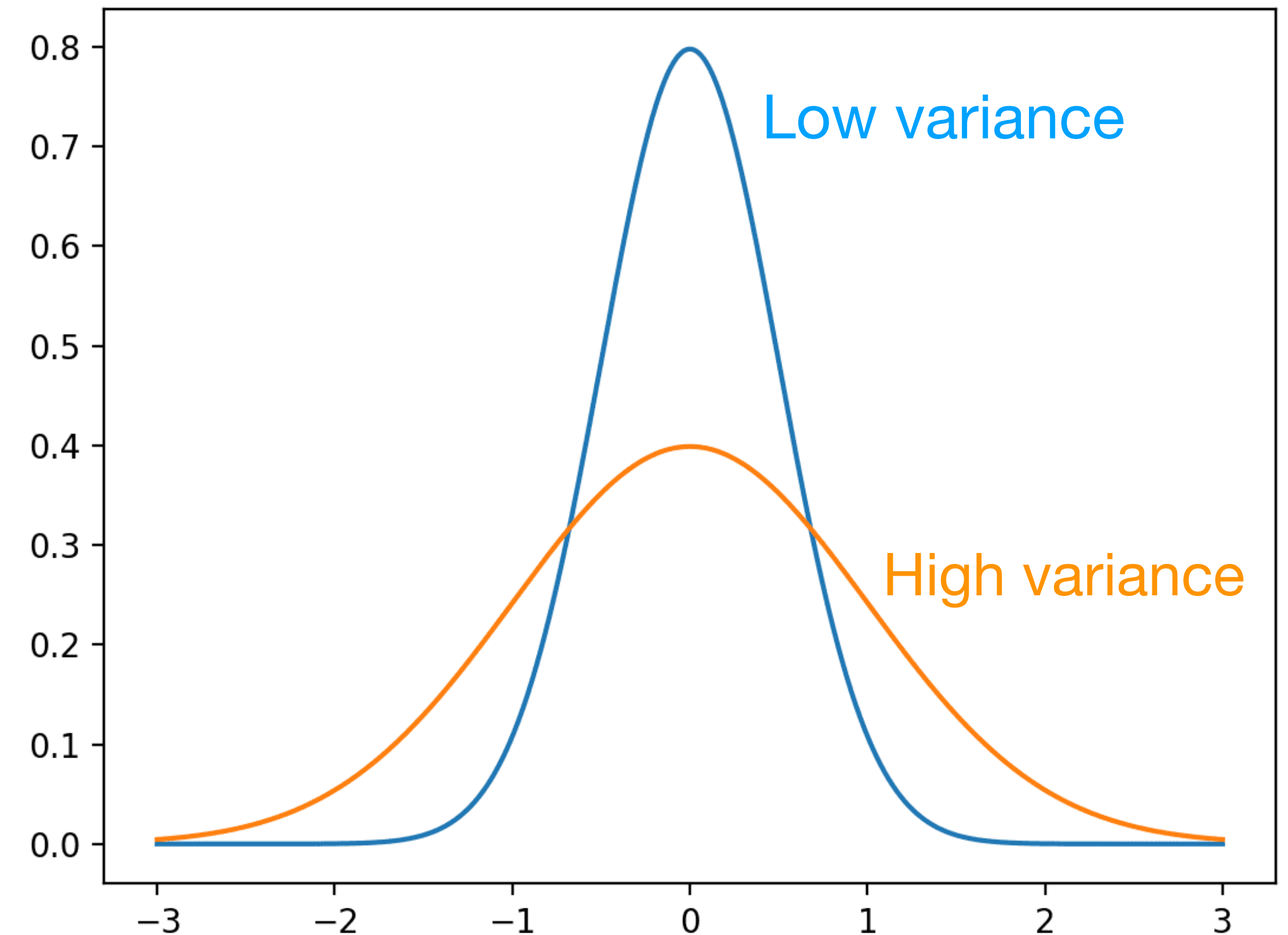
$$E(X_1 + X_2) = E(X_1) + E(X_2)$$

$$E(Y) = aE(X) + b \quad \text{for } Y = aX + b$$

VARIANCE

- Variance of X measures how spread out distribution of X is.

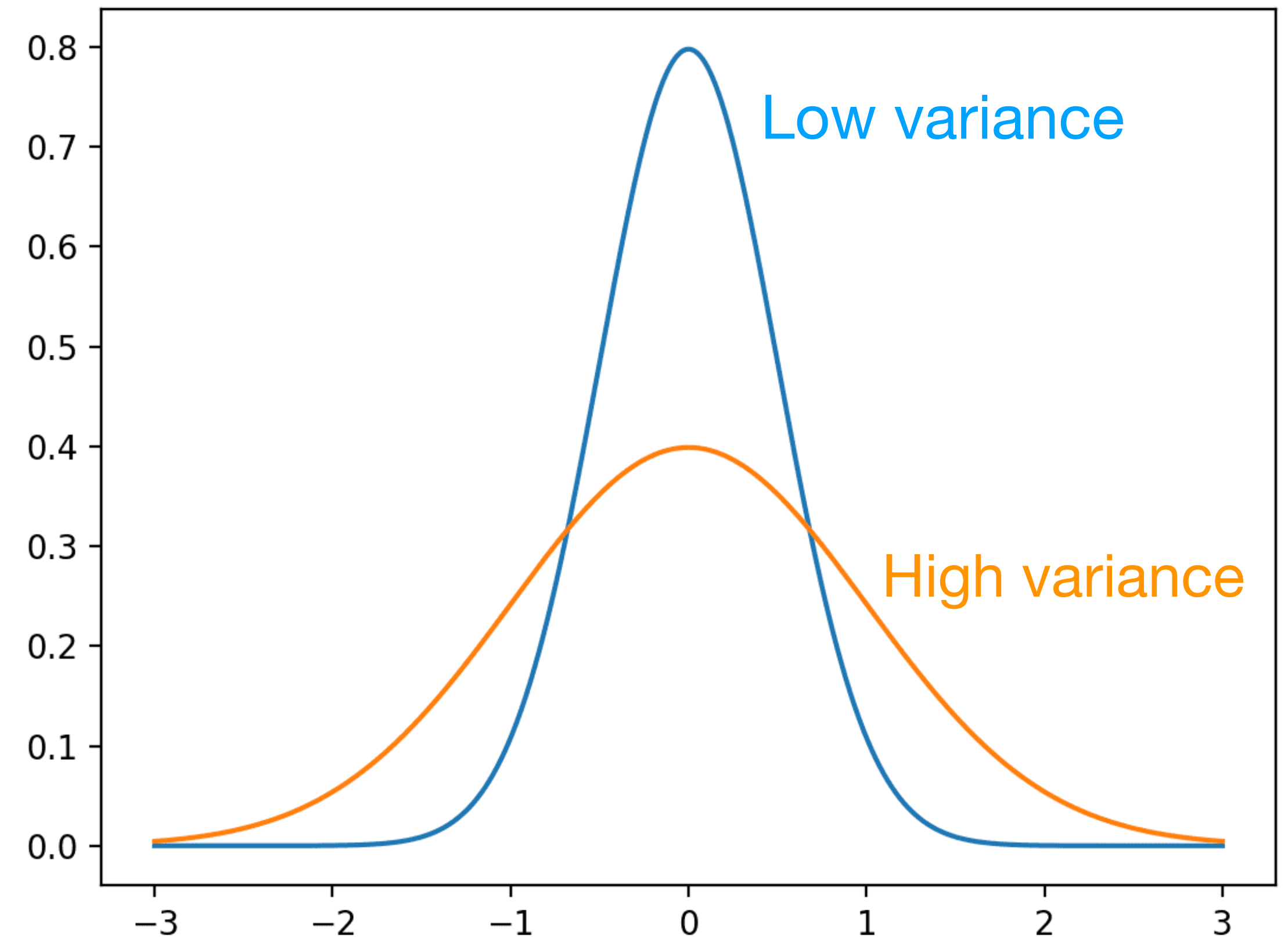
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- Related measure: Standard deviation of $X = \sigma_x = \sqrt{\text{Var}(X)}$

VARIANCE

Ex: Variance of X : outcome of fair six-sided die?

$$E(X) = \frac{7}{2}$$

$$Var(X) = E[(X - E(x))^2]$$

Note that $E[r(X)] = \sum_x r(x)f(x)$

$$= \sum_{x=1}^{x=6} \left(x - \frac{7}{2}\right)^2 \frac{1}{6}$$

$$= \frac{1}{6}((-5/2)^2 + (-3/2)^2 + (-1/2)^2 + (1/2)^2 + (3/2)^2 + (5/2)^2)$$

$$\approx 2.92$$

VARIANCE

- An equivalent formula for variance:

$$Var(X) = E(X^2) - [E(X)]^2$$

Proof:

$$\begin{aligned} Var(X) &= E[(X - \mu)^2] = E[X^2 - 2X\mu + \mu^2] \\ &= E(X^2) - E(2X\mu) + E(\mu^2) \\ &= E(X^2) - 2\mu E(X) + \mu^2 = E(X^2) - 2\mu^2 + \mu^2 \\ &= E(X^2) - \mu^2 \end{aligned}$$

- More useful in calculations.

VARIANCE

Ex: Variance of X : outcome of fair n -sided die.

Helpful formula: $\sum_{x=1}^n x = n(n+1)/2$, $\sum_{x=1}^n x^2 = n(n+1)(2n+1)/6$

$$\begin{aligned} \text{Var}(X) &= E(X^2) - [E(X)]^2 = \sum_{x=1}^n x^2 \frac{1}{n} - \left(\sum_{x=1}^n x \frac{1}{n} \right)^2 \\ &= \frac{1}{n} \times \frac{n(n+1)(2n+1)}{6} - \left(\frac{1}{n} \times \frac{n(n+1)}{2} \right)^2 = \frac{n^2 - 1}{12} \end{aligned}$$

PROPERTIES OF VARIANCE

- Let $Y = aX + b$ for constants a, b .

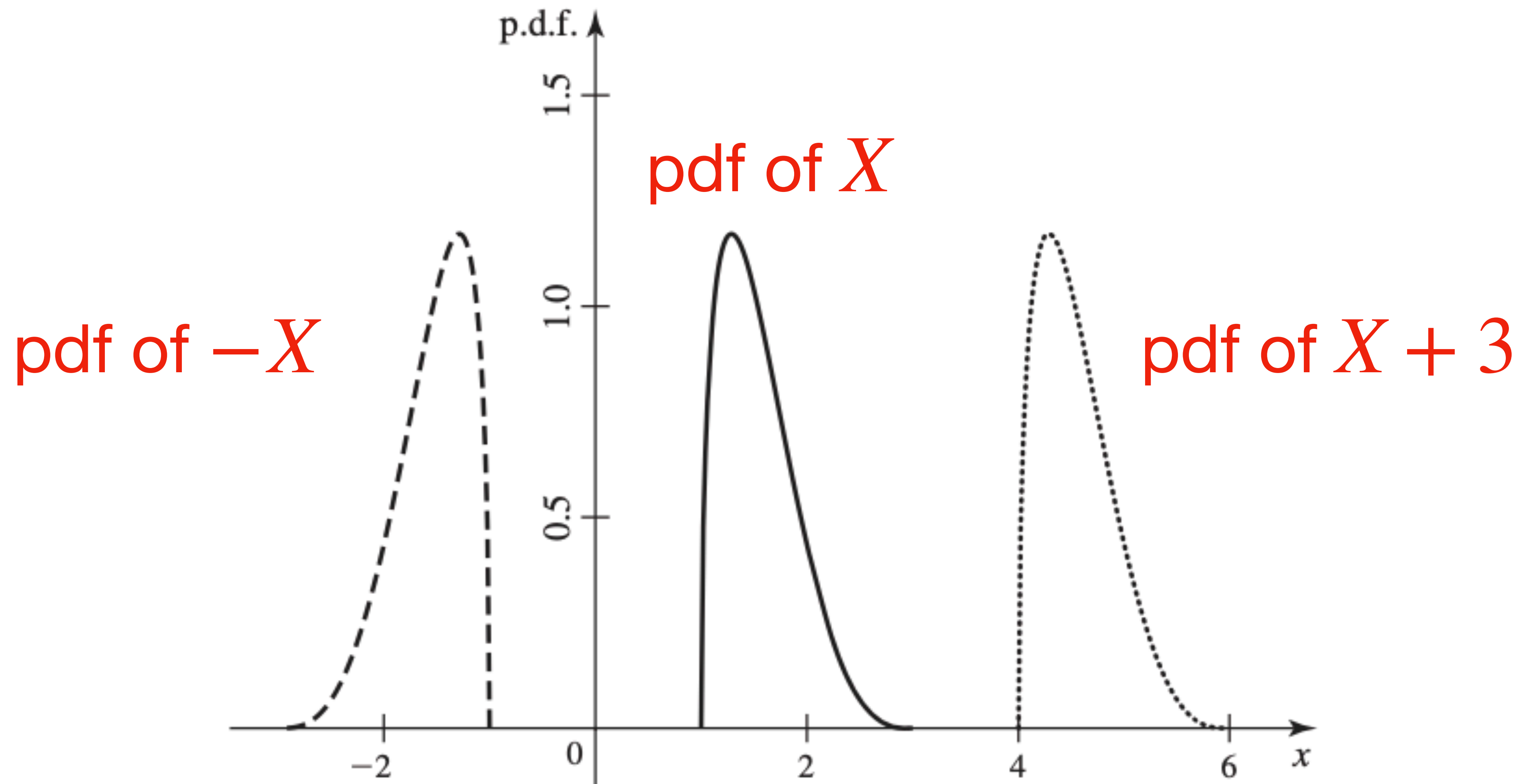
$$\text{Var}(Y) = a^2 \text{Var}(X)$$

Proof:

$$\begin{aligned}\text{Var}(Y) &= E[(aX + b - E(aX + b))^2] \\ &= E[(aX + b - aE(X) - b)^2] \\ &= E[(aX - aE(X))^2] \\ &= E[a^2(X - E(X))^2] = a^2 E[(X - E(X))^2] = a^2 \text{Var}(X)\end{aligned}$$

PROPERTIES OF VARIANCE

- Previous result implies $\text{Var}(X + b) = \text{Var}(X)$, for constant b .



SUMMARIES OF ASSOCIATION BETWEEN TWO VARIABLES

- **Covariance** of X, Y : numerical measure of the degree to which X and Y vary together. Let $E(X) = \mu_x, E(Y) = \mu_y$.

$$Cov(X, Y) = E[(X - \mu_x)(Y - \mu_y)]$$

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- Equivalent formula: $Cov(X, Y) = E(XY) - \mu_x\mu_y$

Proof: $Cov(X, Y) = E[(X - \mu_x)(Y - \mu_y)]$

$$= E(XY) - \mu_y E(X) - \mu_x E(Y) + \mu_x \mu_y$$

$$= E(XY) - 2\mu_x \mu_y + \mu_x \mu_y = E(XY) - \mu_x \mu_y$$

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BUT

is also influenced by the overall magnitudes of X, Y .

$$\text{Cov}(2X, Y) = 2\text{Cov}(X, Y) \text{ for instance.}$$

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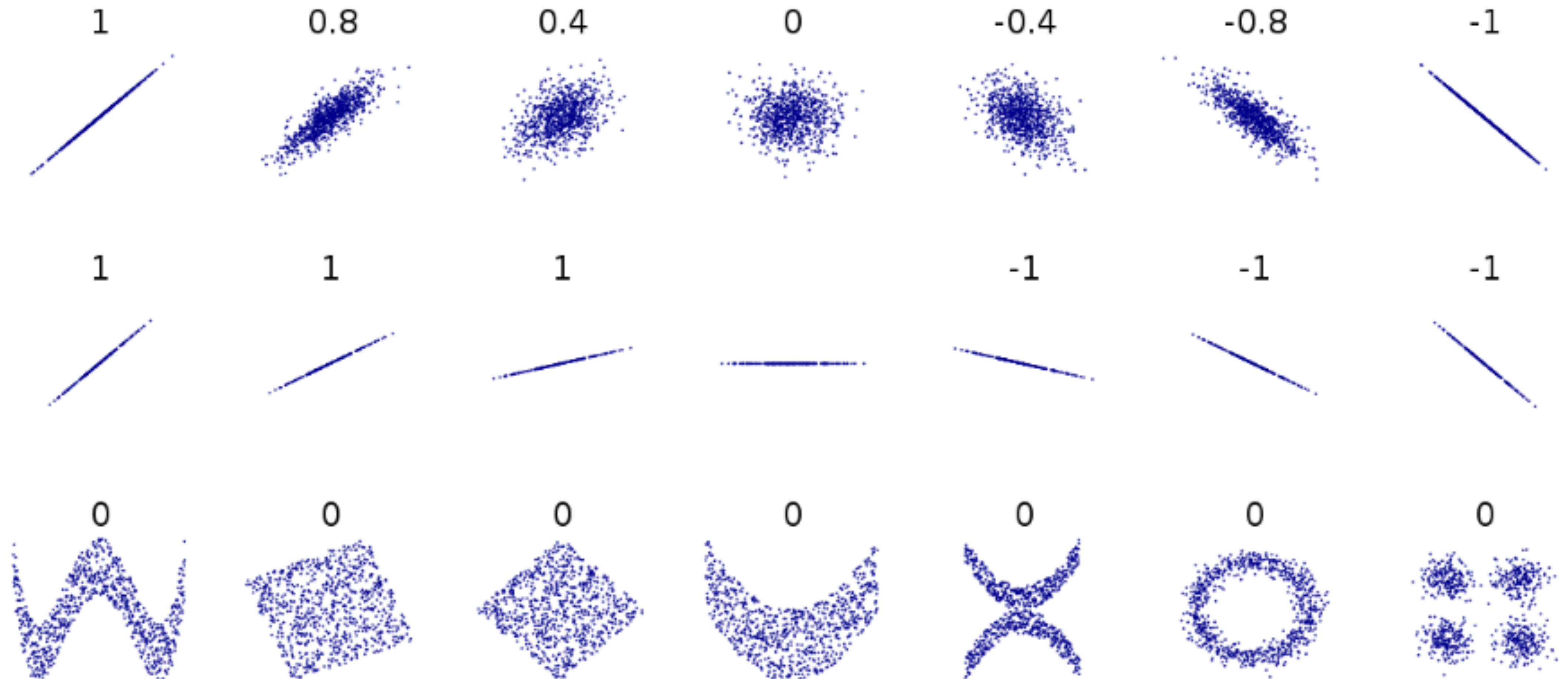
- Better measure, independent of arbitrary changes in scales:

$$\text{Correlation of } X, Y = \rho(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

Measures linear association of X, Y . Always between -1 and 1 .

SUMMARIES OF ASSOCIATION BETWEEN TWO VARIABLES

- Example instances of $\rho(X, Y)$:



SUMMARIES OF ASSOCIATION BETWEEN TWO VARIABLES

- **Note:** Correlation does not imply causation!

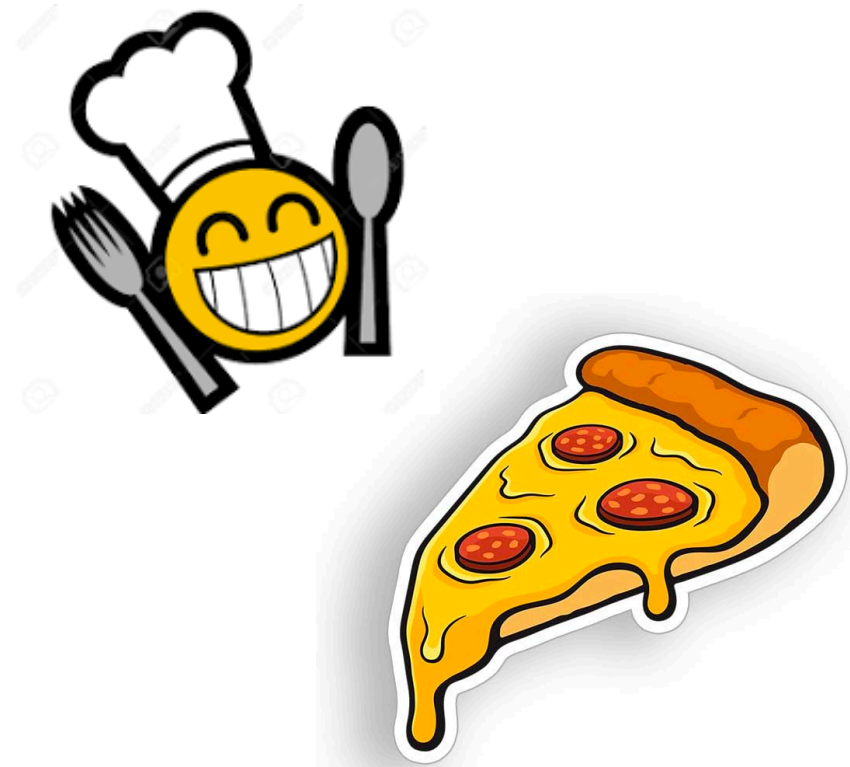
Assume strong ρ between X , Y . Possible scenarios:

- X causes Y .
- Y causes X .
- Z (another factor/set of factors) causes changes in X , Y .

Am I happy because I ate more pizza?

Am I eating more pizza because I am happy?

Am I happy and eating more pizza because prices went down?



VARIANCE OF SUM OF TWO VARIABLES

- For expectation we had: $E(X + Y) = E(X) + E(Y)$.

Does it hold for variance? NO!

- $Var(X + Y) = Var(X) + Var(Y) + 2Cov(X, Y)$

VARIANCE OF SUM OF TWO VARIABLES

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Does it hold for variance? NO!

- $Var(X + Y) = Var(X) + Var(Y) + 2Cov(X, Y)$

Proof: We know $E(X + Y) = \mu_X + \mu_Y$

$$Var(X + Y) = E[(X + Y - E(X + Y))^2] = E[(X + Y - \mu_X - \mu_Y)^2]$$

$$= E[(X - \mu_X)^2 + (Y - \mu_Y)^2 + 2(X - \mu_X)(Y - \mu_Y)]$$

$$= E[(X - \mu_X)^2] + E[(Y - \mu_Y)^2] + 2E[(X - \mu_X)(Y - \mu_Y)]$$

INDEPENDENT RANDOM VARIABLES

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If X, Y independent random variables $E(XY) = E(X)E(Y)$

Note: Generalizes to n independent variables.

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If X, Y independent random variables $E(XY) = E(X)E(Y)$

Proof: (Only for discrete case. Continuous case is similar.)

$$\begin{aligned} E(XY) &= \sum_{\forall x, y} xyf(x, y) = \sum_{\forall x} \sum_{\forall y} xyf(x)f(y) = \sum_{\forall x} xf(x) \sum_{\forall y} yf(y) \\ &= \sum_{\forall x} xf(x)\mu_y = \mu_y \sum_{\forall x} xf(x) = \mu_y\mu_x \end{aligned}$$

$X \perp Y \Rightarrow f(x, y) = f(x)f(y)$

Note: Generalizes to n independent variables.

INDEPENDENT RANDOM VARIABLES

Ex: Roll 2 fair dice. X : outcome of die 1, Y : outcome of die 2.

Mean of their product?

$$X, Y \text{ independent} \Rightarrow E(XY) = E(X)E(Y) = \frac{7}{2} \times \frac{7}{2} = \frac{49}{4}$$